

Daily Logic Drill #1

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Statements & Quantifiers ($\forall/\exists$) · Negating Compound Statements · Negating Quantified Statements · Order of Quantifiers.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Rapid formal logic: evaluate truth conditions, negations of nested predicates, and quantifier order.

1. Negate the nested statement: “ $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : x^2 + y^2 < 1$.”

2. Over the positive integers $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$, determine the truth value of:

$$\forall n \in \mathbb{Z}^+, \exists m \in \mathbb{Z}^+ : m < n$$

3. State the negation of the implication: “If n is even, then n is a multiple of 4 or n is a multiple of 6.”

4. Over the real numbers $\mathbb{R}$, determine the truth value of:

$$\exists x \in \mathbb{R}, \forall y \in \mathbb{R} : xy = 0$$

5. Over the non-zero real numbers $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$, determine the truth value of:

$$\forall y \in \mathbb{R}^*, \exists x \in \mathbb{R}^* : xy = 1$$

6. Given propositions P and Q , if the biconditional $P \iff Q$ is false, what is the truth value of $(P \wedge \neg Q) \vee (\neg P \wedge Q)$?

7. Negate the condition: $x \in [-2, 5] \setminus \{0\}$.

8. Determine the complete set of real values of k for which the statement is true:

$$\forall x \in \mathbb{R}, (x < 0 \implies x^2 > k)$$

9. If the conditional statement $(P \wedge Q) \implies (R \vee S)$ is false, state the unique truth assignment for the propositions P, Q, R, S .

10. Negate: “For every even integer $n > 2$, there exist primes p and q such that $n = p + q$.”

Section B Manipulation Drills

(≤ 50 s each)

Multi-premise deductions, quantifier set characterisations, and exactly-one-true logic.

1. An integer $n > 1$ is defined to be an $\mathcal{S}$ -number if every factor $d > 1$ of n is a multiple of some element of $\mathcal{S} = \{2, 3\}$. Which of the following is the correct definition of an integer $n > 1$ that is *not* an $\mathcal{S}$ -number? (after TMUA Specimen Paper 2 Q17)

- A) Every factor $d > 1$ of n is not a multiple of 2 and not a multiple of 3.
- B) There exists a factor $d > 1$ of n such that $\gcd(d, 6) = 1$.
- C) There exists a factor $d > 1$ of n that is neither 2 nor 3.
- D) No factor $d > 1$ of n is a multiple of 6.

2. Let P, Q, R be propositions. Suppose you are given that the compound statement

$$(P \implies Q) \wedge (Q \implies R) \wedge (R \implies P)$$

is false, but **exactly two** of the three implications are true. Can P, Q , and R all have the same truth value? (after TMUA 2020 Paper 2 Q20)

3. Let S be the set of real numbers x such that:

$$\forall y \in (0, 1), \quad xy < x + y$$

Determine the set S in interval notation. (after TMUA 2022 Paper 2 Q13)

4. Negate the statement: “ $\forall x > 0, \exists y > 0 : (y < x \wedge xy \geq 1)$.” Is the original statement true or false over $\mathbb{R}$?

5. Three statements about an integer n are given:

- I. n is a multiple of 6.
- II. n is odd.
- III. n is a prime strictly greater than 3.

Given that **exactly one** of I, II, III is true, which statement is impossible to be the true one? (after TMUA 2016 Paper 2 Q4)

6. Negate the nested statement: “ $\exists p$ prime such that $\forall q$ prime with $q > p$, q is odd.” Is the original statement true or false?

7. Consider the claim: “ $\forall a, b, c \in \mathbb{Z}^+, (a \mid bc \implies (a \mid b \vee a \mid c))$.” (a) State the precise negation of this claim. (b) Find a counterexample with the minimal value of a . (after TMUA 2021 Paper 2 Q4)

8. Given that the compound implication $((P \vee Q) \implies R)$ is false, how many of the 8 possible truth assignments for (P, Q, R) are consistent with this fact?

9. Negate the statement: “ $\forall x \in \mathbb{R}, (x \in \mathbb{Q} \vee (x \notin \mathbb{Q} \wedge x > 0))$.” Give a numerical witness proving the negation is true.

10. Negate: “For every prime p , $p^2 + 2$ is prime.” What is the smallest prime counterexample to this statement? (after TMUA 2022 Paper 2 Q3)

Section C Substitution & Structure

(≤90 s each)

Multi-step negations, nested quantifiers, and your first ordering-of-quantifiers MCQ.

- For real x and positive integer n , consider: I. For all x and for all n , $x^2 < n$. II. For all x , there exists n such that $x^2 < n$. III. There exists x such that for all n , $x^2 < n$. Which of I, II, III are true? (after TMUA 2022 Paper 2 Q10)
- Determine the complete set of real values of k for which the following is true: “For all real x , if $x < k$, then $x^2 > k$.”
 - no real numbers
 - $k < 0$
 - $k \leq 0$
 - $k \leq 1$
 - all real numbers
 - $k > 0$
- Which is the correct negation of “For all primes p , p is odd or $p = 2$ ”?
 - There exists a prime p such that p is odd or $p = 2$.
 - There exists a prime p such that p is not odd and $p \neq 2$.
 - For all primes p , p is not odd and $p \neq 2$.
 - There exists a prime p such that p is not odd or $p \neq 2$.
- Negate: “There exists a positive integer n such that $n^2 + n + 1$ is even.” Is the negation true or false?
- Negate: “There exists a positive integer n such that for every positive integer $k \leq 10$, k divides n .” Is the original statement true or false?
- Consider the statement about a real number x : “There exists a real number y such that $xy - x + y$ is positive.” For how many real values of x is this true? (after TMUA 2022 Paper 2 Q13)
 - no values of x
 - exactly one value of x
 - all except exactly one value of x
 - all except exactly two values of x
 - all values of x
- Negate: “There exists a real number x such that for every real number y , $xy = y$.” Is the original statement true or false?

8. Negate: “For every positive integer n , there exists a positive integer $m < n$ that is prime.” Consider the cases $n = 1$ and $n = 2$ carefully. Is the original statement true or false?

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

- Consider the statement: “For every finite sequence of positive integers, there exists a positive integer m such that every term of the sequence is at most m .” Which of the following is the correct negation?
 - There exists a finite sequence of positive integers such that, for every positive integer m , some term exceeds m .
 - There exists a finite sequence such that, for every positive integer m , every term is at most m .
 - For every finite sequence, there exists a positive integer m such that some term exceeds m .
 - There exists an infinite sequence such that no positive integer bounds all terms.
- Determine, with brief justification, whether the following is true or false: “For every positive real number ε , there exists a positive integer N such that for every integer $n \geq N$, $\frac{1}{n} < \varepsilon$.”
- A selection S of n terms is taken from the arithmetic sequence $2, 5, 8, 11, \dots, 92$. What is the smallest value of n for which S must contain two distinct terms summing to 94? (*after TMUA 2022 Paper 2 Q8*)
 - 15
 - 16
 - 17
 - 18
 - 24
 - 31
- Statement (*): “For every finite set of primes $\{p_1, \dots, p_k\}$, there exists a prime not in the set.”
 - Negate (*).
 - Is (*) true or false? Justify briefly.
- Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Consider: I. “For every x , there exists y such that $f(y) = x$.” II. “There exists y such that for every x , $f(y) = x$.”
 - Are I and II logically equivalent?
 - Give a concrete function f for which I is true but II is false.

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at [github.com/David Akinyele-Aje/speed-maths](https://github.com/David-Akinyele-Aje/speed-maths)